

Element F: Consideration of Design Viability

"Design viability" is the capacity of the proposed solution to solve the problem. Design viability refers to how your design decisions are tethered to your design requirements. For example, if one of your design requirements is that the total cost of the prototype cannot exceed X, then you could present a table of costs for each component of your proposed design solution as justification for why your design is likely to meet that requirement.

"Credible evidence" means it shouldn't just be your hunch that this will work. Be creative about how you validate the viability of your design. You should go beyond just having a stakeholder or expert reviewer say that the design seems viable. For example, you could point to evidence that a sub-system already functions, or you demo a proof of concept, or you've discussed with an expert that said, "I've seen something similar work in this other related situation...", etc.

Use as many rows as needed. Remember, this chart below may help you organize your ideas, but you will need to actually execute the "credible evidence."

Name:

- T.A.G. (Tracking And Guardian)

Description:

- T.A.G. is a tracker used to efficiently, effectively, and successfully track from your device. With the help of Microbit, we designed a device that can track family members and loved ones and older people with disabilities. This device can track real-time location allowing the guardian or caretaker to keep track of the person using the device. T.A.G. allows features like SOS buttons for emergency situations and sets boundaries so if the person using the device goes out of that boundary, that guardian or caretaker would be notified. This tracker provides a peace of mind and well being of the people who use it.

Main Focus:

- The main focus of T.A.G is to ensure the safety of the person using the device by providing their live location and using other features to protect them.

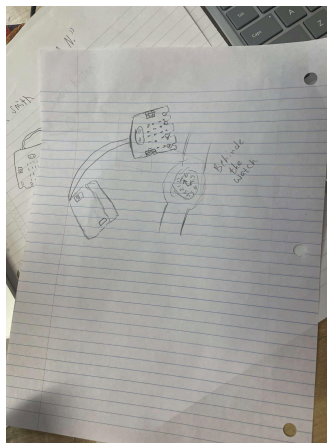
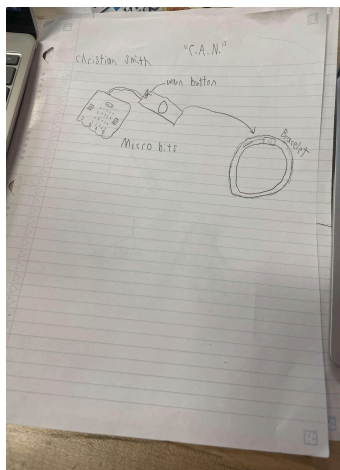
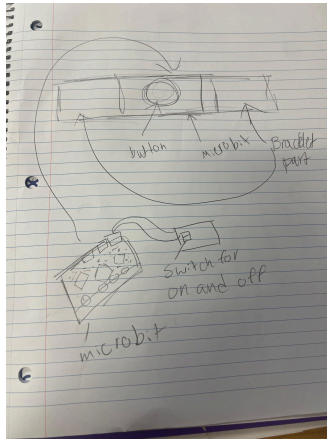
Functions:

- Tracking(From Guardians device)
- Notifies guardians in situations where there is suspicious activity
- SOS button
- Boundary settings
- Send signals to guardians device

Price:

- \$40-60

Images:



Stakeholders:

- Officer O'briant - The OPS school officer.

(Helped make major changes to change the main focus of the project. Change the tracking from children being kidnaped to just an all around tracker.)

- School Teacher - Every OPS teacher in the school.

(Mr. Roberson has helped us come up with ideas and move forward with the project. After the form was filled out by some of the teachers, we changed some of our focuses and they gave us input on what we should do with our project)

- Students Parents - The Parents of all the students in our engineering class.

(They did nothing, thanks parents.)